

MeSA 2.0 — Technical Design & 10-Week Implementation Plan

Project: Medication & Safety Assistant Robot (MeSA 2.0) **Owner:** Vrishaank Mishra

Target: Working tabletop prototype in 10 weeks (~June 15 – August 23, 2026) **Effort**

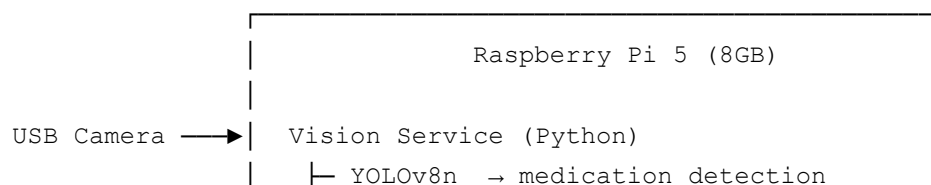
budget: 10–15 hrs/week (~120–150 hrs total) **Repo:** `VrishaankMishra/mesa-ai-robot`
(Layer 1 software already scaffolded: YOLOv8 detection, Streamlit dashboard, pyttsx3 voice, unit tests, simulation loop)

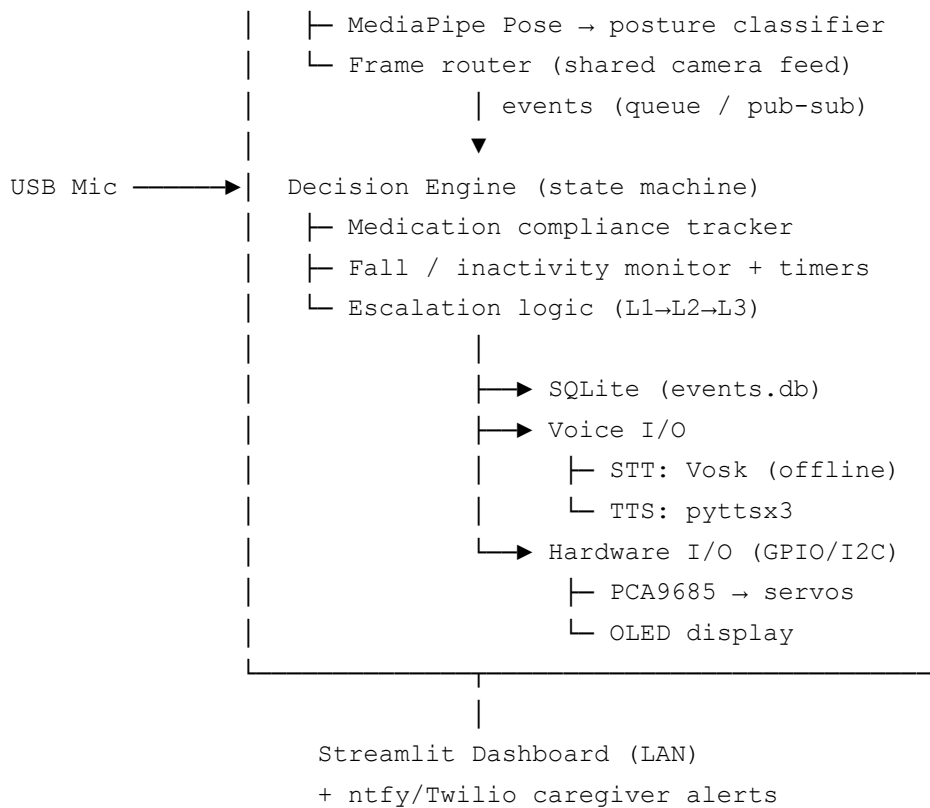
1. Prototype Definition of Done

The prototype is “working” when, in a single live demo on the Raspberry Pi 5:

1. **MED-DEMO:** Camera identifies 5+ medication bottles with $\geq 90\%$ top-1 accuracy at 0.5–1.5 m, and flags an unknown bottle as “wrong medication.”
 2. **LOG-DEMO:** Removing and returning a bottle creates a timestamped “taken” event in SQLite, visible on the Streamlit dashboard.
 3. **FALL-DEMO:** System classifies standing / sitting / lying in real time; lying >30 s triggers a spoken check-in.
 4. **VOICE-DEMO:** Four voice commands work end-to-end (next med, did-I-take, call for help, date/time) with spoken responses.
 5. **ESCALATE-DEMO:** No-movement timeout triggers the 3-level escalation (ask → notify → caregiver alert).
 6. (*Stretch*) Servo head turns toward detected person; OLED shows emotion states.
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2. System Architecture





Key design decisions

Decision	Choice	Rationale
Detection model	YOLOv8n (nano)	Already in repo; runs 5–10 FPS on Pi 5 CPU; fine-tune on Colab free tier
Pose model	MediaPipe Pose (lite)	CPU-friendly, no training needed; posture from keypoint geometry rules
STT	Vosk small-en model	Fully offline, ~50MB, real-time on Pi; Whisper-tiny as fallback if accuracy poor
TTS	pyttsx3	Already integrated; offline
IPC between services	Python multiprocessing.Queue (single process tree)	Avoids MQTT/network complexity at this scale
Storage	SQLite	Already chosen; zero-admin
Alerts (L2/L3)	ntfy.sh push notifications (free), Twilio SMS optional	No server to build
Dev	Build on laptop first, deploy	Faster iteration; Pi only constrains

workflow	to Pi weekly	performance
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Concurrency model: one main process, three worker threads/processes — (1) vision loop, (2) audio loop, (3) decision engine consuming an event queue. Frame rate target: detection every 3rd frame; pose every frame at 640×480.

3. Hardware (Bill of Materials)

Core (Weeks 1–6, ~\$165–235)

#	Item	Spec	Est. Cost	Needed by
H1	Raspberry Pi 5	8GB	\$80	Week 5
H2	Pi 5 power supply	27W USB-C official	\$12	Week 5
H3	microSD card	64GB+ A2, flashed Pi OS 64-bit	\$15	Week 5
H4	Active cooler for Pi 5	Official	\$5	Week 5
H5	USB webcam	Logitech C920 (1080p)	\$50	Week 1
H6	USB microphone	Mini USB or lapel-style	\$20	Week 4
H7	Speaker	USB or 3.5mm powered	\$15	Week 4

Robotics extras (Weeks 8–9, ~\$50)

#	Item	Spec	Est. Cost
H8	Servos ×2	SG90 or MG90S	\$15
H9	Servo driver	PCA9685 (I2C, 16-ch)	\$10
H10	Pan-tilt bracket kit	For camera/head	\$15
H11	OLED display	SSD1306 128×64 I2C	\$10

Procurement notes: Webcam + mic + speaker first (laptop dev needs them). Pi 5

bundle ordered by end of Week 3 (Micro Center St. Davids, PA or online) so it arrives before Week 5. Servos/OLED ordered Week 6.

Test props

5–10 medicine bottles (Tylenol, Vitamin D, multivitamin, ibuprofen, etc.), a “medication station” tray with fixed camera mount position, masking tape for floor test zone.

4. Software Stack

Layer	Technology	Notes
Language	Python 3.11	Same version laptop + Pi
Vision	OpenCV, Ultralytics YOLOv8, MediaPipe	YOLO fine-tuned on custom dataset
Dataset tooling	Roboflow (free), Google Colab (free GPU)	Annotation + training
Audio	Vosk (STT), pyttsx3 (TTS), sounddevice/pyaudio	All offline
Data	SQLite via <code>sqlite3</code> <code>stdlib</code>	Tables: <code>events</code> , <code>medications</code> , <code>schedule</code>
Dashboard	Streamlit	Reuse existing dashboard, extend
Alerts	<code>requests</code> → <code>ntfy.sh</code> ; optional Twilio	
Hardware I/O	<code>gpiozero</code> , <code>adafruit-circuitpython-pca9685</code> , <code>adafruit-circuitpython-ssd1306</code>	Pi only
Testing	<code>pytest</code> (unit), recorded-video fixtures (integration)	Already have unit tests
Version control	Git + GitHub, feature branches, PR-style merges	Issues board = ticket tracker
Packaging	<code>requirements.txt</code> + <code>requirements-pi.txt</code> ; systemd service on Pi	Auto-start on boot

5. Skills Required (and where they're learned in-plan)

Skill	Level needed	Learned during
Python (functions, classes, threading, queues)	Intermediate	Throughout; threading in Week 6
Dataset collection & annotation (Roboflow)	Beginner→Intermediate	Week 1–2
Training/fine-tuning YOLOv8 on Colab	Beginner→Intermediate	Week 2
Evaluating CV models (precision/recall, confusion matrix)	Beginner	Week 2–3
MediaPipe pose keypoints + geometric rules	Beginner→Intermediate	Week 5
State machines & event-driven design	Beginner→Intermediate	Week 3, 6
SQLite schema design + queries	Beginner	Week 3
Audio I/O, STT/TTS pipelines	Beginner	Week 4
Linux/Raspberry Pi admin (SSH, systemd, I2C)	Beginner	Week 5, 8
Servo control via PWM/I2C	Beginner	Week 8
Git workflow, GitHub Issues, README/docs	Intermediate	Throughout
Demo design & technical writing	Intermediate	Week 10

6. 10-Week Timeline with Milestones & Tickets

Tickets follow `AREA-###` IDs. Each has acceptance criteria (AC) and an hour estimate. Create them as GitHub Issues with labels `vision`, `voice`, `engine`, `hw`, `infra`, `docs`. Weekly budget $\approx$ 12 hrs.

Week 1 (Jun 15–21) — Dataset & Dev Environment

Milestone M1: Annotated dataset v1 uploaded to Roboflow

ID	Ticket	AC	Hrs
INFRA-001	Set up dev environment on laptop	Repo cloned; venv with pinned <code>requirements.txt</code> ; existing unit tests pass	2
INFRA-002	Create GitHub Issues board with all tickets	All tickets entered with labels + milestones	1
VIS-001	Build capture rig + photo protocol	Fixed camera position documented; lighting/angle/distance checklist written	2
VIS-002	Capture 100 images × 6 bottles	≥600 images, 3 lighting conditions, 3 angles, 3 distances; includes cluttered backgrounds	5
VIS-003	Annotate dataset in Roboflow	All images boxed + labeled; 70/20/10 train/val/test split; augmentations configured	3

Week 2 (Jun 22–28) — Train & Evaluate Detector

Milestone M2: Fine-tuned YOLOv8n hitting ≥90% mAP@50 on val set

ID	Ticket	AC	Hrs
VIS-004	Train YOLOv8n on Colab	Training notebook in repo; best.pt exported; ≥90% mAP@50	4
VIS-005	Evaluation report	Confusion matrix + per-class precision/recall committed to <code>/docs</code> ; failure cases listed	2
VIS-006	Live inference script	<code>detect_live.py</code> shows boxes + confidence at ≥5 FPS on laptop; "unknown bottle" path when confidence < threshold	4
VIS-007	Hard-negative round	50+ images of failure cases added; retrained if mAP < 90%	2

Week 3 (Jun 29–Jul 5) — Compliance Tracking & Database

Milestone M3: Bottle removed/returned → event logged in SQLite

ID	Ticket	AC	Hrs
ENG-001	SQLite schema + data access module	<code>events</code> , <code>medications</code> , <code>schedule</code> tables; CRUD functions; pytest coverage	3
ENG-002	Presence state machine per bottle	States: present → absent → returned; debounced (3 s) so hand occlusion doesn't false-trigger	4
ENG-003	"Medication taken" event rule	Absent ≥10 s then returned <code>taken</code> event with timestamp + med name in DB	2
DASH-001	Extend Streamlit dashboard	Today's meds, taken/missed status, event history table reading live from SQLite	3

 **Order Pi 5 bundle (H1–H4) this week.**

Week 4 (Jul 6–12) — Voice Assistant

Milestone M4: 4 voice commands working end-to-end on laptop

ID	Ticket	AC	Hrs
VOX-001	Mic capture + Vosk STT loop	Continuous transcription; wake word "MeSA" gates commands	4
VOX-002	Intent parser	Keyword/regex intents: NEXT_MED, DID_I_TAKE, HELP, DATE_TIME; unit-tested on 20 phrasings each	3
VOX-003	Intent → DB → spoken answer	"Did I take Vitamin D?" answers correctly from real events table via pytsx3	3
VOX-004	HELP intent → alert stub	"Call for help" sends ntfy.sh push notification	2

Week 5 (Jul 13–19) — Fall Detection + Pi Bring-up

Milestone M5: Standing/sitting/lying classified live; Pi 5 running the detector

ID	Ticket	AC	Hrs
VIS-008	MediaPipe Pose integration	33 keypoints rendered live at ≥ 10 FPS (laptop)	2
VIS-009	Posture classifier	Rules on torso angle + hip/shoulder height classify standing/sitting/lying with $\geq 90\%$ accuracy on a 50-clip recorded test set	5
VIS-010	Lying-duration trigger	Lying > 30 s fires <code>possible_fall</code> event; spoken "Are you okay?" check-in	2
HW-001	Pi 5 setup	Pi OS 64-bit, SSH, camera working, repo deployed, YOLO+pose FPS benchmarked and recorded	3

 **Safety note:** test fall detection with staged poses on cushions only; document that this is an assistive aid, not a medical device.

Week 6 (Jul 20–26) — Decision Engine Integration

Milestone M6: All subsystems running together from one entry point (laptop)

ID	Ticket	AC	Hrs
ENG-004	Event bus + main orchestrator	<code>main.py</code> launches vision, audio, engine threads; all events flow through one queue	5
ENG-005	Escalation state machine	L1 spoken check-in $\rightarrow$ no response 60 s $\rightarrow$ L2 ntfy push $\rightarrow$ no response $\rightarrow$ L3 caregiver alert (SMS or repeated push); fully unit-tested with simulated time	4
ENG-006	Inactivity monitor	No person/motion for configurable window (default 1 h) triggers escalation L1	2
INFRA-003	Config file	All thresholds (timeouts, confidence, schedule) in one <code>config.yaml</code>	1



Order servos, PCA9685, pan-tilt kit, OLED (H8–H11).

Week 7 (Jul 27–Aug 2) — Full System on Raspberry Pi

Milestone M7: End-to-end demo runs on the Pi unattended for 2+ hours

ID	Ticket	AC	Hrs
HW-002	Port full stack to Pi	All Week-6 functionality on Pi; <code>requirements-pi.txt</code> ; perf tuning (frame skipping, resolution) to keep pose ≥ 5 FPS	5
HW-003	systemd service	MeSA auto-starts on boot; logs to file; survives reboot	2
ENG-007	Soak test + bug bash	2-hour continuous run; no crashes; memory stable; issues filed and fixed	4
DASH-002	Dashboard reachable on LAN	Phone/laptop on same Wi-Fi can view dashboard served from Pi	1

Week 8 (Aug 3–9) — Robotic Personality

Milestone M8: Servo head tracks person; OLED shows emotion states

ID	Ticket	AC	Hrs
HW-004	PCA9685 + pan-tilt assembly	Camera mounted on pan-tilt; servos sweep via script	3
HW-005	Face/person tracking head	Head turns to keep detected person centered (simple proportional control); no jitter/oscillation	4
HW-006	OLED emotion display	😊 idle, ⚠ alert, ❤ greeting states driven by engine events	3
INFRA-004	Demo video #1	60–90 s clip of med detection + fall check-in recorded for portfolio	1

(If hardware is delayed, Week 8 swaps with Week 9.)

Week 9 (Aug 10–16) — Accuracy Hardening & Stretch Goals

Milestone M9: Demo-grade reliability; one stretch goal landed

ID	Ticket	AC	Hrs
VIS-011	Detector robustness pass	Test at demo location lighting; retrain/threshold-tune until MED-DEMO criteria met	4
VOX-005	Voice robustness pass	90%+ command success at 1–2 m in normal room noise; mis-recognition fallback ("Sorry, say again?")	3
STRETCH-001	Pick ONE: schedule generator (CSV prescription list → reminder schedule) OR family dashboard auth view OR face recognition greeting	Chosen feature demo-able	5

Week 10 (Aug 17–23) — Demo, Docs & Showcase Package

Milestone M10: Working prototype + complete portfolio package

ID	Ticket	AC	Hrs
DOC-001	README overhaul	Architecture diagram, setup guide, demo GIFs, results table (mAP, posture accuracy, FPS)	3
DOC-002	Final demo video	3–4 min: all 5 Definition-of-Done demos shown live, narrated	3
DOC-003	Project abstract + poster draft	1-page abstract (problem, system, results, impact) for science fair / applications	3
ENG-008	Dress rehearsal	Full scripted demo run twice without intervention; fallback recorded video ready	2

7. Milestone Summary

Week	Milestone	Proof
1	M1 Dataset v1	Roboflow project link
2	M2 Detector $\geq 90\%$ mAP@50	Eval report in repo
3	M3 Compliance logging	Dashboard shows live "taken" event
4	M4 Voice commands $\times 4$	Screen recording
5	M5 Fall detection + Pi alive	Posture demo clip; Pi FPS benchmark
6	M6 Integrated system (laptop)	One-command launch, all events flowing
7	M7 Full stack on Pi, 2 h soak	systemd logs
8	M8 Servo head + OLED	Tracking demo clip
9	M9 Hardened + 1 stretch goal	Robustness test results
10	M10 Showcase package	Video, README, abstract

8. Risks & Mitigations

Risk	Likelihood	Mitigation
Pi 5 too slow for YOLO + pose simultaneously	Medium	Frame skipping, 640×480, YOLOv8n only; fallback: run detection on med-station ROI only
Pi shipping delay	Medium	All Weeks 1–4 and 6 are laptop-only; Pi needed truly from Week 7
Vosk accuracy poor	Medium	Swap to whisper-tiny (still offline) — interface is abstracted behind one STT module
Dataset overfits to home lighting	High	Hard-negative round (VIS-007) + Week 9 hardening at demo location
Scope creep (stretch goals)	High	Stretch only in Week 9, only one, only if M7 done
Servo/I2C wiring issues eat time	Medium	Personality features isolated to Week 8; project demos fine without them

9. Weekly Operating Rhythm

- **Start of week (30 min):** review milestone, pick tickets, move to "In Progress."
- **Each session:** commit to a feature branch; close tickets with the AC checklist.
- **End of week (30 min):** record a 30-second progress clip (these become the portfolio montage), write 3-line progress log in `docs/journal.md` — admissions-gold later.